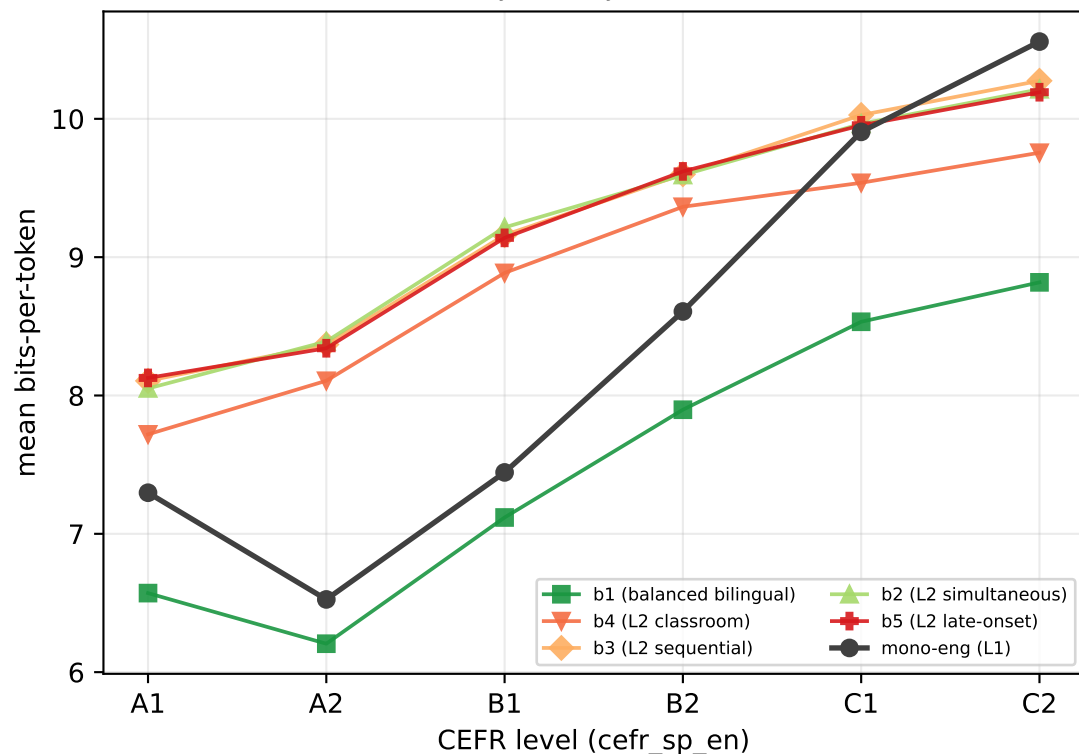


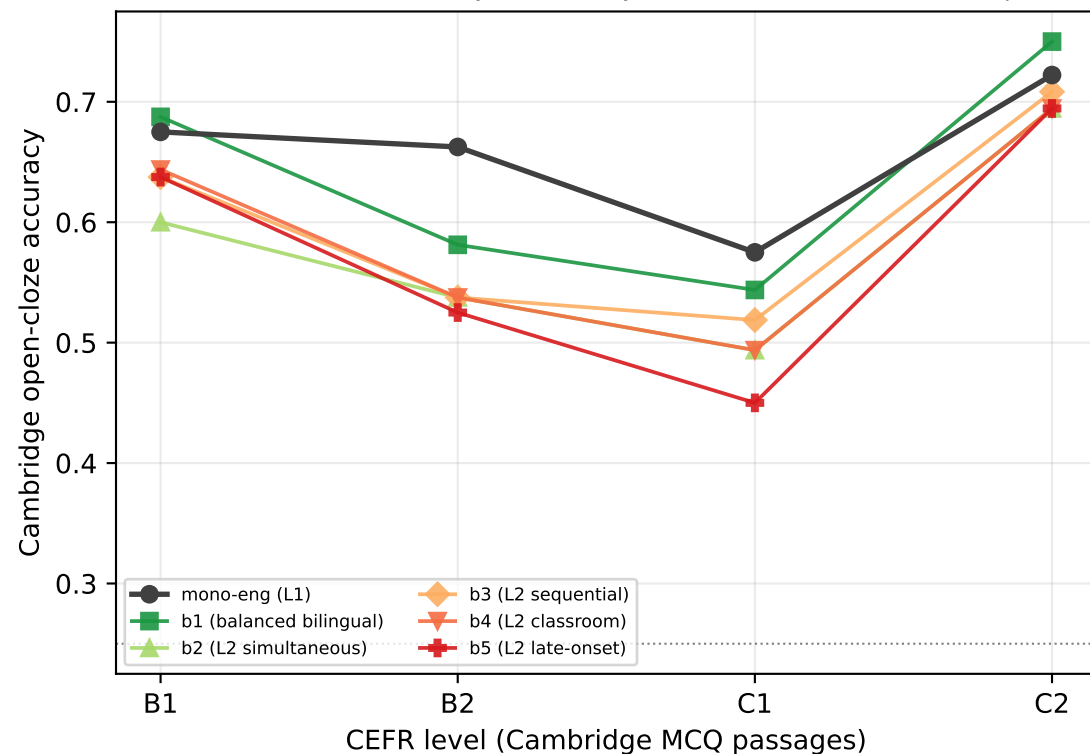
Beetle 125M as a real-student simulator — six convergent signals.

Every metric independently confirms a canonical L2-learner property: graded difficulty (A,B), learner-error preference (C), L1-transfer (D), CEFR-ordered vocabulary acquisition (E), and L2-error ranking (F).

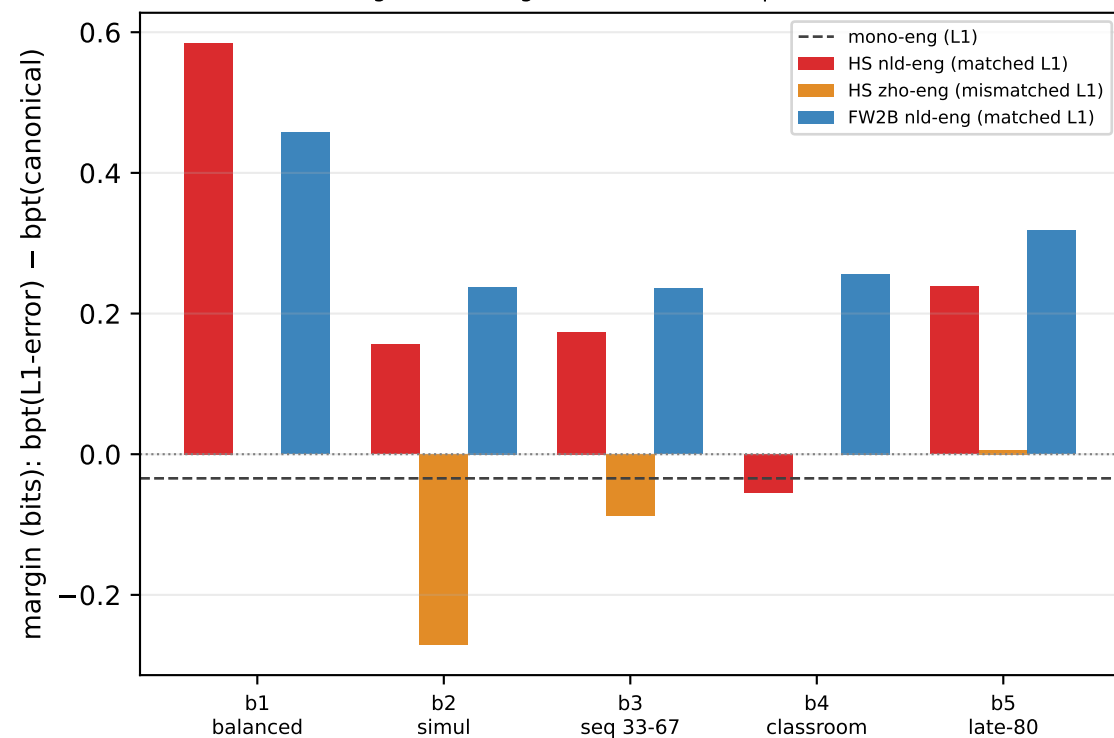
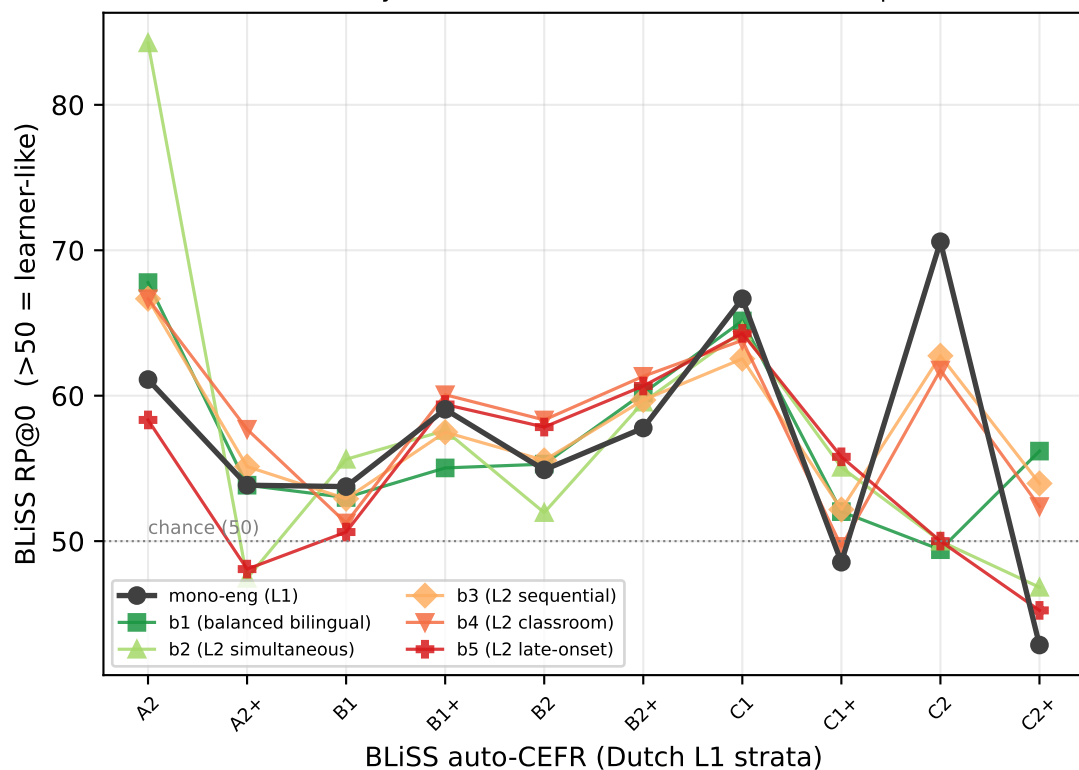
(A) §4.4 Graded text difficulty — every Beetle model is monotonic A1→C2



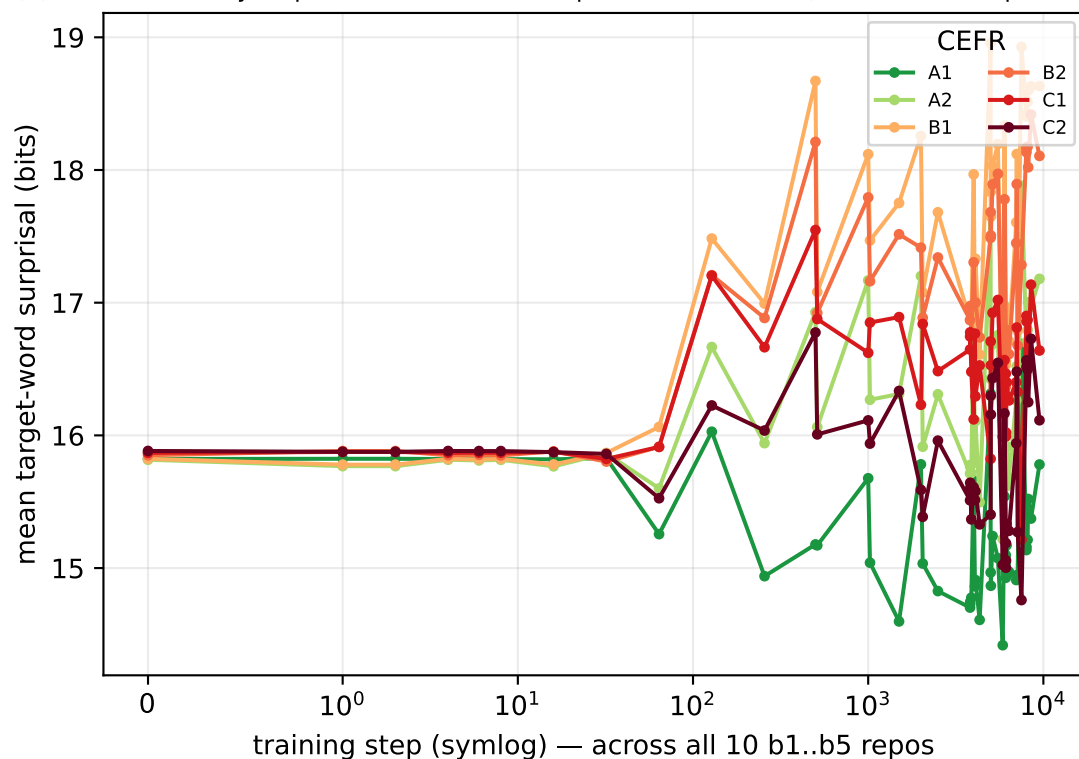
(B) §4.1 Graded cloze difficulty — accuracy declines B1→C1 (real student pattern)



(C) BLISS — every Beetle model >chance on learner-error preference (D) §4.2 L1-transfer simulation — matched bilingual disambiguates; mismatched produces the canonical L1-transfer error (negative margin)



(E) §4.3 Vocabulary acquisition — A1 words drop first, then A2..C2 (canonical L2 acquisition order)



(F) §4.2 JFLEG — every Beetle model ranks corrected > learner-error (frac > 50%)

